

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

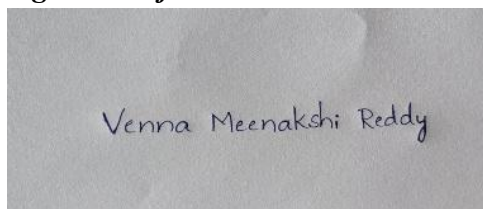
Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

I VENNA MEENAKSHI REDDY(192471048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a piece of paper with the name "Venna Meenakshi Reddy" handwritten in blue ink.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. Define SOAP.

- SOAP stands for Simple Object Access Protocol.
- It is a protocol used for communication between applications.
- It mainly uses XML to exchange data.
- It is commonly used in web services and enterprise applications.

2. Define WSDL.

- WSDL stands for Web Services Description Language.
- It is an XML-based language.
- It describes the functions provided by a web service.
- It specifies the operations, messages, and endpoint of the service.

3. What is an API?

- API stands for Application Programming Interface.
- It allows two software applications to communicate.
- It defines rules and methods for accessing data or services.
- APIs are commonly used in web and mobile applications.

4. Define XML Schema.

- XML Schema is also called XSD (XML Schema Definition).
- It defines the structure of an XML document.
- It specifies elements, attributes, and data types.
- It helps validate XML data.

5. What is a Web Service?

- A web service allows applications to communicate over a network.
- It enables data exchange between different systems.
- It can use technologies such as SOAP and REST.
- It is commonly accessed through the internet using HTTP.

6. Define REST.

- REST stands for Representational State Transfer.
- It is an architectural style for designing web services.
- It uses HTTP methods such as GET, POST, PUT, and DELETE.
- REST commonly uses JSON for data exchange.

7. What is a Protocol?

- A protocol is a set of rules for communication.
- It defines how data is sent and received.
- It helps devices and applications communicate correctly.
- Examples include HTTP, HTTPS, TCP, and FTP.

8. Define Endpoint in Web Services.

- An endpoint is a specific address used to access a web service.
- It is usually represented by a URL.
- It identifies a particular service or resource.
- For example, /api/users can be an API endpoint.

9. What is a Service in Web Applications?

- A service is a software component that provides a specific function.
- It can be accessed by other applications.
- It performs a particular task or operation.
- Examples include login, payment, and weather services.

10. What is the difference between SOAP and REST?

- SOAP is a protocol, while REST is an architectural style.
- SOAP mainly uses XML, while REST commonly uses JSON.
- SOAP is more complex, while REST is simpler and lightweight.
- REST is widely used for modern web and mobile APIs.

11. What is the role of WSDL in Web Services?

- WSDL describes a web service and its functions.
- It specifies the operations available in the service.
- It defines the messages and data types used.
- It provides information about the service endpoint.

12. What is JSON and how is it used in APIs?

- JSON stands for JavaScript Object Notation.
- It is a lightweight format for exchanging data.
- It is easy for humans and computers to read.
- It is commonly used in REST APIs.

13. What is HTTP and why is it used in Web Services?

- HTTP stands for HyperText Transfer Protocol.
- It is used to transfer data between clients and servers.
- Web services use HTTP to send requests and responses.
- Common HTTP methods are GET, POST, PUT, and DELETE.

14. What is meant by Request and Response in Web Communication?

- A request is a message sent by the client to the server.
- It asks the server to perform an operation or provide data.
- A response is a message sent by the server to the client.
- It contains the requested data or result.

15. What is the importance of XML in Web Services?

- XML provides a structured way to represent data.
- It is platform and language independent.
- It allows different applications to exchange data.
- It is widely used in SOAP-based web services.

16. What is the purpose of an API Endpoint?

- An API endpoint is a specific URL of an API.
- It provides access to a particular resource or operation.
- Clients send requests to the endpoint.
- The server returns an appropriate response.

17. What is the difference between HTTP and HTTPS?

- HTTP is used for normal web communication.
- HTTPS is the secure version of HTTP.
- HTTPS encrypts data using TLS.
- HTTPS protects sensitive information during transmission.

18. What is meant by RESTful Services?

- RESTful services follow the principles of REST.
- They treat data as resources.
- Resources are identified using URLs.
- They use HTTP methods such as GET, POST, PUT, and DELETE.

19. What is a Namespace in XML?

- An XML namespace uniquely identifies XML elements.
- It prevents naming conflicts between different XML documents.
- Namespaces are declared using the xmlns attribute.
- They are useful when combining different XML vocabularies.

20. What is the role of a Client and Server in Web Services?

- The client sends requests to the server.
- The server receives and processes the requests.
- The server sends a response back to the client.
- This communication forms the basic client-server model.